

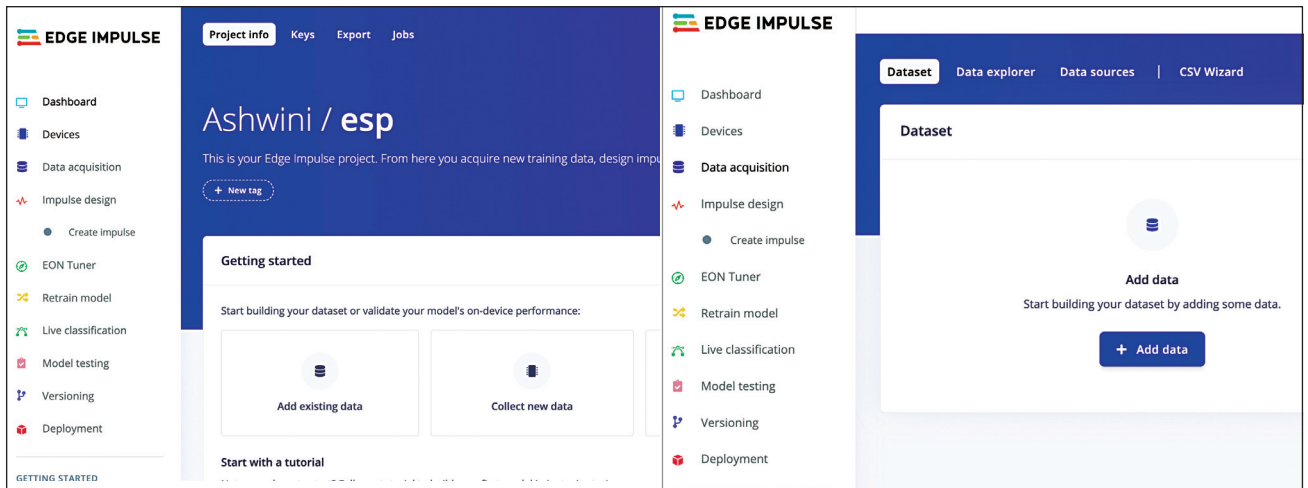


Fig. 3: Edge Impulse device addition and data collection



Fig. 4: QR code to test and run the ML model

tons from various angles, which is crucial for real-world deployment. Edge Impulse also allows connecting your cell phone or laptop for input device convenience in the data acquisition phase. Refer to Fig. 3 for Edge Impulse device addition and data collection.

The project

The Edge Impulse project is broadly divided into the following steps:

(a) Data acquisition. This involves collecting various types of data such as images, sound, temperatures, distances, etc. Some of these data are separated as test data, while all others are used as training data.

(b) Impulse design. This step is further subdivided into creating impulses, with sub-divisions for:

1. *Input parameters:* image [width, height], sound [sound parameter].

2. *Processing block:* How to pro-

cess the input data

3. *Learning block:* [Object data of this model]

4. *Image processing:* Generate features of the collected images

5. *Object detection:* Select your neural network model and train the model

The object detection part requires expertise, or one could call it a trial-and-error effort, to achieve an accuracy level of 85% or above. There are several models to try, and anything above 90% is considered excellent. However, it should not be 100% accurate, as this could indicate issues with the data. For this device, the accuracy achieved was 98.6%, which is commendable for a starter project, considering the limited dataset of around 40 instances. Refer to Fig. 3 for more details.

Model testing

You can test your model on the test data first. See how it works out there, and then point your device to the real-life data to observe its performance! On the browser page, explore this feature. Fig. 4 displays a QR code for testing and running the ML model.

This feature is available in the dashboard of the Edge Impulse opening page. You can scan the image

on your mobile and run the model there, or you can run it directly in the browser. Point the camera at the buttons and check whether it can count them. Fig. 5 showcases the data acquisition of three samples.

ML deployment

To run the model on a Raspberry Pi computer, you have to download the *.eim file onto the Raspberry Pi computer. However, unlike other hardware such as Arduino, Nicla Vision, and ESP32, where you can download directly, for Raspberry Pi you need to install Edge Impulse on the Raspberry Pi computer first. Inside the edge-impulse-daemon software, download this file. Don't worry; Edge Impulse has a dedicated page for installing Edge Impulse on Raspberry Pi. (Link: <https://docs.edgeimpulse.com/docs/development-platforms/officially-supported-cpu-gpu-targets/raspberry-pi-4>)

Now that Edge Impulse is installed on the Raspberry Pi computer, run edge-impulse-linux-runner. Note: Keep the internet on for the Raspberry Pi. On a Raspberry Pi computer terminal, connect it to your Edge Impulse page (run edge-impulse-linux-runner - clean to switch projects). This command will automatically compile and down-